

CUSTOMER INSIGHTS & SALES PERFORMANCE DASHBOARD FOR A RETAIL BUSINESS

CAPSTONE PROJECT – END-TO-END DATA ANALYTICS FOR BUSINESS GROWTH

PRESENTED BY:

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PROJECT UNDER:

IIT KANPUR CERTIFIED DATA ANALYTICS PROGRAM

POWERED BY:

K-TECH LEARN & PROSPER

Business Problem Understanding

The retail company faced several business challenges affecting growth and profitability.

Major Problems:

- Declining repeat customers
- Uneven sales across regions
- Low-performing product categories
- Lack of centralized dashboard for decision-making

Project Goal:

To analyze sales, customer behavior, and product performance data to generate actionable business insights and improve decision-making.

Dataset Overview

Dataset Name:

Sample Superstore Dataset

Dataset Size:

- 9,994 rows
- 21 columns

Key Features:

- Sales
- Profit
- Customer Segment
- Product Category
- Region
- Order Date

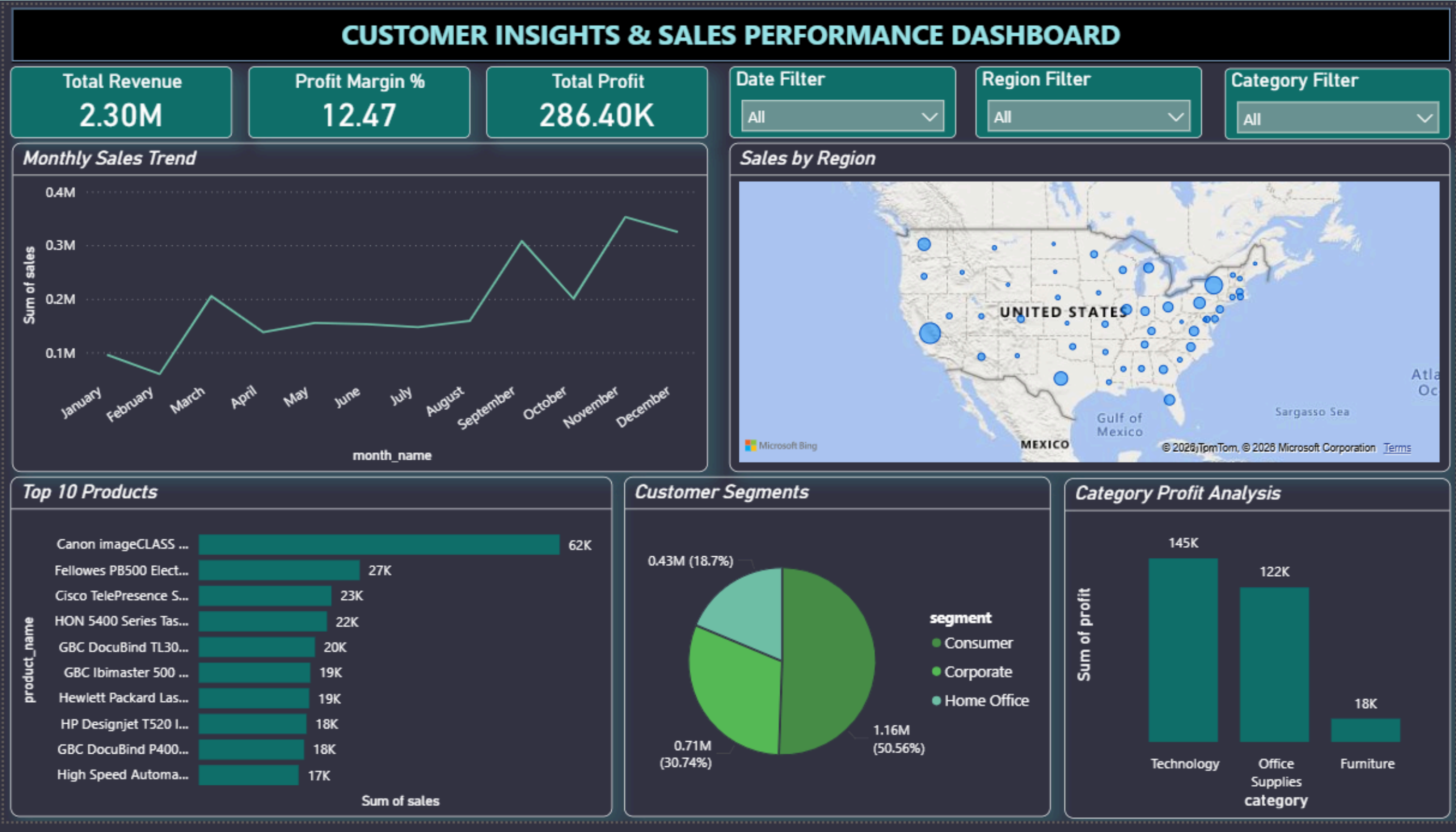
Tools Used:

- Python (Pandas, Matplotlib, Scikit-learn)
- Power BI
- Google Colab

Key Insights Found

- Consumer segment generated highest sales revenue.
- Sales peaked during November and December.
- Technology category produced highest profit.
- Some regions contributed significantly higher sales.
- High-performing products should be prioritized for inventory planning.

Interactive Dashboard Demo

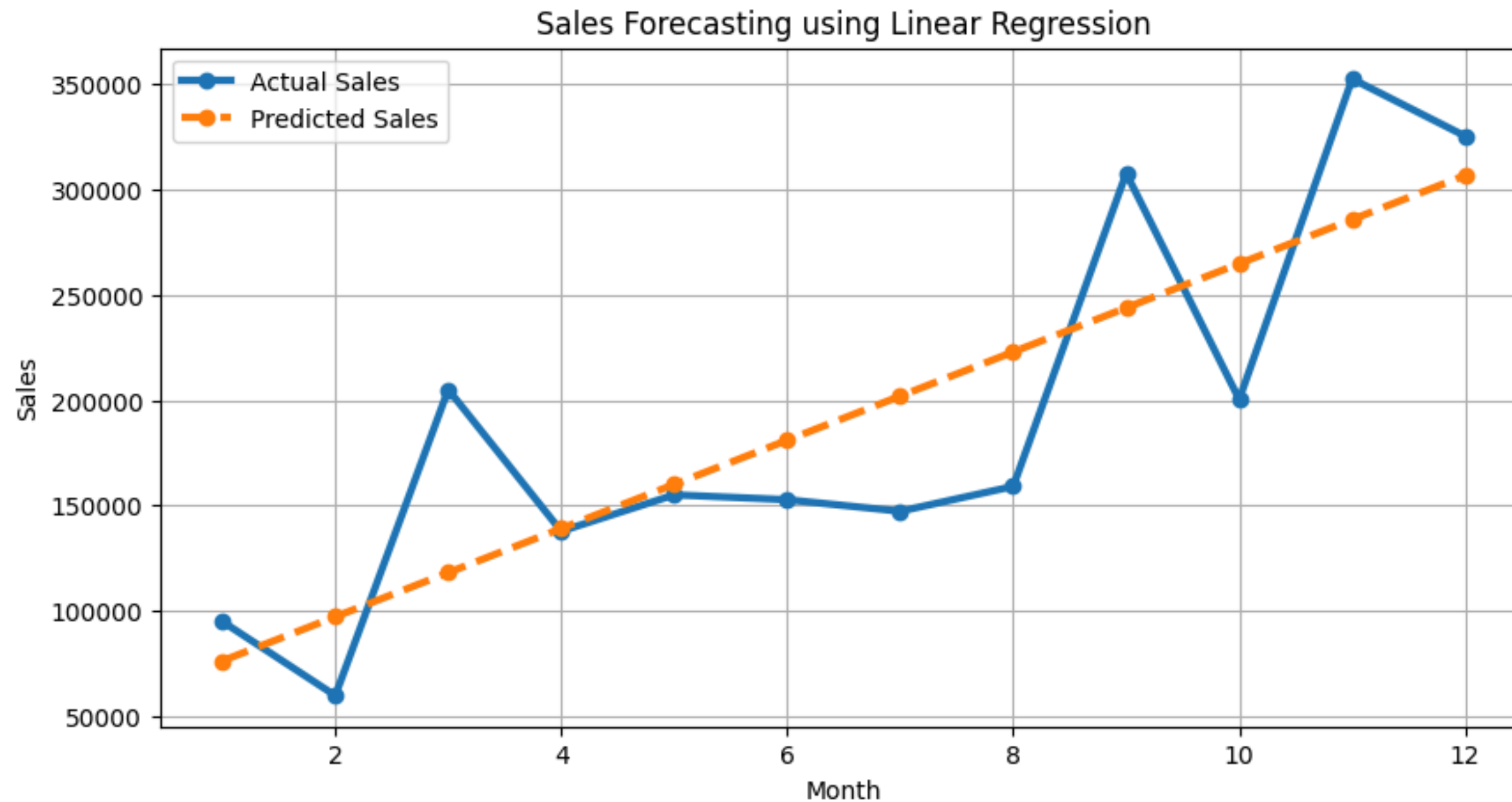


Developed an interactive Power BI dashboard to visualize:

- Revenue & Profit KPIs
- Monthly sales trends
- Top-selling products
- Customer segmentation
- Region-wise sales performance
- Interactive filters

Predictive Analytics

Sales Forecasting using Linear Regression



A Linear Regression model was used to forecast future sales trends using monthly sales data.

Model Performance:

- R^2 Score: 0.67

Outcome:

- The model successfully identified future sales patterns and seasonal growth trends.
- Forecasting can help businesses improve inventory planning and sales strategies.

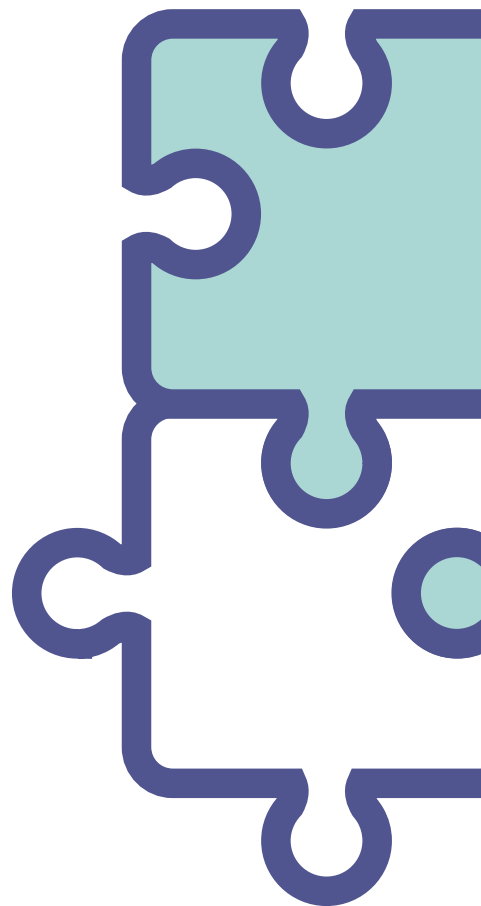
Business Recommendations

- Increase marketing efforts in high-performing regions.
- Launch loyalty programs to improve repeat purchases.
- Focus inventory on high-demand products.
- Reduce focus on low-profit product categories.
- Use dashboards for faster business decision-making.

ROI Impact of Recommendations

Expected Business Impact:

- Improved customer retention
- Increased regional sales performance
- Better inventory optimization
- Higher profitability from high-margin products
- Faster and data-driven management decisions



Thank you!